

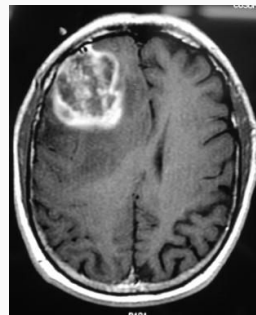
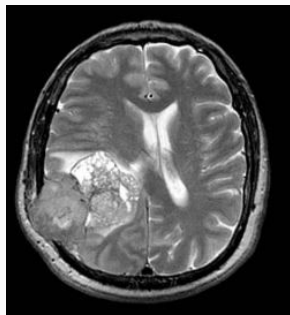
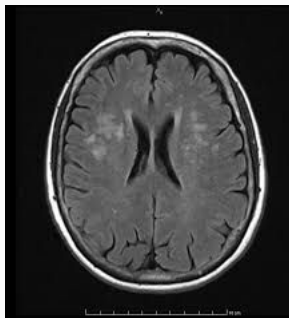
Neural Networks for Tumor Detection and Classification from MRI

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Summary

This project will focus on evaluating existing architectures for image classification and the criteria for selecting an appropriate model. After selection, a pretrained model will be modified and retrained to detect and classify tumors from MRI scans.



Purpose & Objectives

- Benefit to society
- Learn CNNs
 - Existing Models
 - Transfer Learning



Potential Models

U-Net - specifically designed for biomedical image segmentation

DenseNet - it performs well even with limited amounts of labeled data

ResNeXt - builds on the proven ResNet architecture

3D CNN (3D U-Net or 3D ResNet) - volumetric data

Data Set

- Kaggle - common data collection for machine learning
- Combination of the following three datasets :
 - Figshare
 - SARTAJ dataset
 - Br35H
- Categories
 - Glioma
 - Meningioma
 - Pituitary
 - No tumor

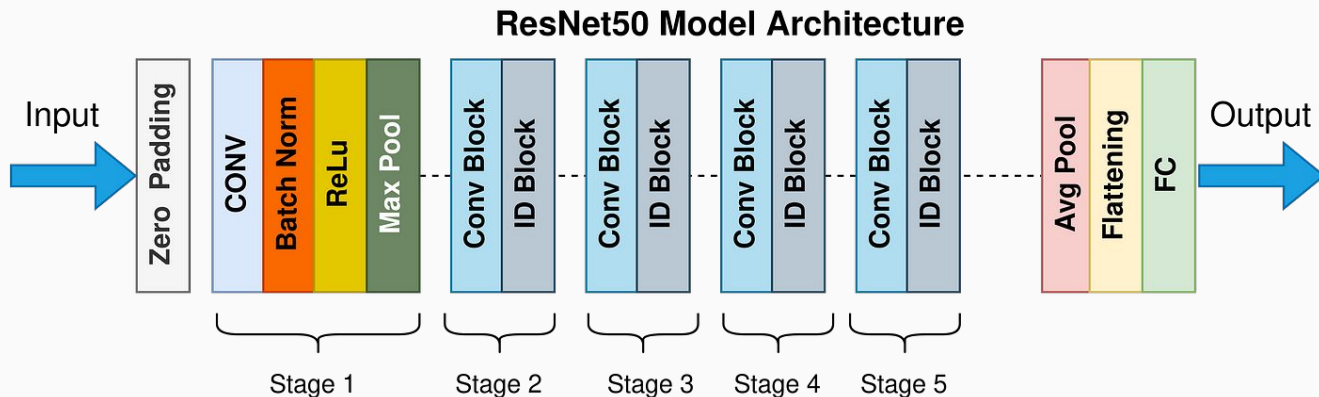
The Kaggle logo, featuring the word "kaggle" in a lowercase, blue, sans-serif font.

Strategy

- Preprocessing
 - MRI data is often high-dimensional
 - normalization, resizing, and augmenting
- Transfer Learning
 - medical imaging datasets are often small

Potential Further Exploration

- Compare multiple models
- Streamline inputs
- Create new architecture



Conclusion

- Challenging task
- Scope

